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Clinicians vs. Artificial Intelligence in Patient Outcome Prediction in the Intensive Care Unit

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ABSTRACT

Accurate prediction of patient outcomes in the intensive care unit (ICU) is critical for clinical decision-making. While artificial intelligence (AI) has shown potential in retrospective prediction, direct comparisons with human clinicians, particularly in prospective real-world settings, remain unclear. We conducted a mixed retrospective and prospective study to compare clinician and AI performances in ICU patient outcome prediction. In the retrospective setting, both clinicians and AI, trained on 46,631 ICU admissions, predicted 990 ICU admissions only based on patient data. In the prospective setting, clinician predictions for 238 ICU admissions were collected during active patient care, whereas AI made predictions for the same admissions using extracted patient data. Target outcomes were in-hospital and 30-day mortality, acute kidney injury, delirium, and ICU and hospital lengths of stay. In the retrospective setting, AI generally outperformed clinicians but aggregated predictions from seven clinicians outperformed AI. In the prospective setting, subspecialized physicians generally outperformed AI, whereas physicians in training and nurses generally underperformed AI. Combinations of clinician and AI predictions often resulted in the best prediction performance. Our findings provide a comprehensive evaluation of clinician and AI performances in ICU outcome prediction under both retrospective and real-world prospective conditions, setting important prediction performance benchmarks.

INTRODUCTION

The intensive care unit (ICU) is a dynamic environment where clinicians make frequent, high-stakes decisions for critically ill patients.¹ Accurate prediction of patient outcomes such as mortality and length of stay (LOS) can support timely interventions, efficient resource allocation, and improved patient care. Although traditional scoring systems such as the Acute Physiology and Chronic Health Evaluation (APACHE)² and Simplified Acute Physiology Score (SAPS)³ are widely used, they offer limited predictive accuracy at the individual level.⁴

The growing volume and complexity of ICU data exceed human cognitive limits, leading to information overload and potentially suboptimal decisions.⁵ Machine learning-based artificial intelligence (AI) offers powerful tools to analyze these data, identify patterns, and generate accurate predictions. Numerous studies have demonstrated strong AI performance in predicting outcomes such as sepsis, delirium, and hospital LOS.⁶

However, few studies have directly compared AI and clinician performances in ICU patient outcome prediction, and most evaluations have been limited to retrospective settings. As a result, it remains unclear whether AI prediction performance holds up in prospective, real-world clinical settings or how it compares with clinician prediction performance.⁷

To address this gap, we compared the predictive performances of clinicians and AI for several ICU patient outcomes using both retrospective and prospective designs.

RESULTS

Patient Cohorts

Table S1 summarizes the patient characteristics of the three cohorts used in this study for AI model training, retrospective comparisons, and prospective comparisons. While baseline characteristics were similar between the model training and retrospective cohorts, the prospective cohort had higher in-hospital mortality and delirium rates, as well as a longer mean hospital LOS. The severity of illness scores also indicate that the prospective cohort was more ill in general.

Tables S2 and S3 compare the baseline characteristics of the patients included and excluded in acute kidney injury (AKI) and delirium modeling, respectively. The included and excluded patients were significantly different for both AKI and delirium. In the AKI cohort, the included patients were generally more severely ill than the excluded, with higher SOFA and APACHE scores and longer LOS. In the delirium cohort, the excluded patients had higher Intensive Care Delirium Screening Checklist (ICDSC)⁸ scores and mortality rates than the included.

Best AI Models

Table S4 lists the features selected for at least one AI model across all outcomes. Tables S5 and S6 report the 5-fold cross-validation prediction performances of all AI models for the classification and regression outcomes, respectively. For all outcomes, XGBoost was the best model based on median area under the receiver operating characteristic curve (AUROC) and root mean squared error (RMSE) for classification and regression, respectively. The performances of the best XGBoost models are comparable to previously reported performances on ABeICU.⁹

Figure S1 shows the calibration plots and Brier scores of the four classification models, based on the test data of the retrospective setting. All classification models were reasonably calibrated, with Brier scores ranging from 0.1145 to 0.1926.

Figures S2-S7 are the SHapley Additive exPlanations¹⁰ (SHAP) beeswarm plots showing the top 19 features of each of the 6 best models. The most important features align with clinical knowledge and what has been reported in the literature (e.g., APACHE scores for mortality, urine output and creatinine for AKI, Glasgow Coma Scale [GCS] scores for delirium and LOS).

The optimized model output thresholds were 0.45, 0.48, 0.52, and 0.5 for in-hospital mortality, 30-day mortality, AKI, and delirium, respectively. These thresholds are close to the neutral threshold of 0.5 despite that threshold selection optimized F1 score.

Figures S8 and S9 report the decision curve analysis results for the classification models in the retrospective and prospective settings, respectively. In all cases except prospective AKI prediction, the AI models were better than treating all or none. Optimized model thresholds were associated with positive net benefits except for retrospective 30-day mortality and prospective AKI prediction.

Retrospective Clinician vs. AI Comparisons

Figure 1 and Table S7 compare the retrospective prediction performances of clinicians and AI for the four classification outcomes. Figure 2 and Table S8 show the corresponding results for the two regression outcomes.

For the classification outcomes, AI generally outperformed clinicians in the one- and two-clinician groups in terms of most performance metrics, with some exceptions such as specificity in the two-clinician group for in-hospital mortality (median: 0.925, 95% confidence interval [CI]: [0.895-0.953] vs. median: 0.896, 95% CI: [0.861-0.926]; $p < 0.001$) or sensitivity in the one-clinician group for AKI (median: 0.652, 95% CI: [0.564-0.738] vs. median: 0.551, 95% CI: [0.464-0.650]; $p < 0.001$). However, in the seven-clinician group, clinicians tended to outperform AI for all outcomes except delirium; AI outperformed clinicians by the largest margin in delirium prediction. Both clinicians and AI were more specific than sensitive for the two mortality outcomes and AKI that exhibited low event rates. Specificity and sensitivity were more balanced for delirium for both clinicians and AI.

For the LOS outcomes, AI outperformed clinicians in general except mean absolute error (MAE) in the two-clinician group for ICU LOS (median: 4.436, 95% CI: [4.109-4.786] vs. median: 4.534, 95% CI: [4.275-4.800]; $p < 0.001$) and MAE in the two-clinician group for hospital LOS (median: 16.73, 95% CI: [15.46-18.08] vs. median: 17.14, 95% CI: [16.18-18.08]; $p < 0.001$). Both clinicians and AI exhibited poor R^2 scores for both LOS outcomes. Unlike the classification outcomes, clinicians did not outperform AI in the seven-clinician group.

Prospective Clinician vs. AI Comparisons

Figure 3 and Table S9 compare the prospective prediction performances of clinicians and AI for the classification outcomes. Figure 4 and Table S10 show the corresponding results for the regression outcomes.

For the classification outcomes, subspecialized physicians generally outperformed AI across all outcomes except delirium, with some exceptions such as sensitivity for in-hospital mortality (median: 0.625, 95% CI: [0.384-0.857] vs. median: 0.700, 95% CI: [0.444-0.923]; $p < 0.001$). For delirium, there was no significant difference between subspecialized physicians and AI in terms of all metrics. The prediction performances of physicians in training and nurses tended to be lower than those of subspecialized physicians but still outperformed AI for 30-day mortality with respect to most performance metrics. AI generally outperformed trainees and nurses in AKI and delirium predictions with some exceptions such as specificity in nurses for delirium (median: 0.681, 95% CI: [0.500-0.850] vs. median: 0.640, 95% CI: [0.444-0.826]; $p < 0.001$).

For both LOS predictions, all clinician types outperformed AI in terms of MAE except for subspecialized clinicians for hospital LOS (median: 42.646, 95% CI: [28.483-61.359] vs. median: 37.798, 95% CI: [25.583-54.494]; $p < 0.001$). AI outperformed clinicians with respect to RMSE and R^2 for both LOS outcomes except for RMSE (median: 9.295, 95% CI: [5.087-13.839] vs. median: 9.641, 95% CI: [6.108-14.277]; $p < 0.001$) and R^2 (median: 0.022, 95% CI: [-0.520-0.317] vs. median: -0.056, 95% CI: [-0.551-0.061]; $p < 0.001$) in trainees for ICU LOS.

Clinician-AI Hybrid Predictions

Tables S11-S14 show the clinician-AI hybrid prediction performance in comparison with the corresponding clinician and AI prediction performances. In both the retrospective and prospective classification tasks, the best sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) within each outcome resulted from a hybrid approach in general; the AND operator leading to the best PPV and specificity, whereas the OR operator yielding the best sensitivity and NPV. The seven-clinician group in the retrospective setting was an exception, where the aggregation of seven clinician predictions performed as well or better than the hybrid predictions.

Similarly, with respect to regression, the hybrid approaches tended to yield the best MAE, RMSE, and R^2 , although AI alone sometimes led to the best performance (e.g., RMSE and R^2 in

the two-clinician group in Table S12). The best hybrid regression performances were also scattered across averaging, minimum, and maximum.

Inter-rater Agreement Analyses

Tables S15-S16 and S17-S18 present the inter-rater agreements between clinicians in the retrospective and prospective settings, respectively. In general, agreements were poor in the retrospective setting, with fair agreements observed in AKI prediction. Agreements were also poor in the prospective setting but there were moderate agreements in the 2-clinician group for in-hospital mortality and 30-day mortality, in the 2- and 4-clinician groups for AKI, and in the 2-clinician group for hospital LOS.

Tables S19-S20 and S21-S22 show the inter-rater agreements between clinicians and AI in the retrospective and prospective settings, respectively. Overall, the retrospective predictions in the 7-clinician group, except AKI, led to moderate agreements but all other results showed poor to fair agreements.

Sensitivity Analysis

Tables S23-S24 report the sensitivity analysis results from adjusting the Likert scale threshold in the retrospective and prospective settings, respectively. As expected from the inherent tradeoffs between performance metrics, grouping the middle Likert scale point into the positive class improved sensitivity and NPV, as well as F1 and balanced accuracy to a lesser extent, at the expense of specificity and PPV. Overall, the 7-clinician group and subspecialized physicians outperformed AI to a lesser extent. The Likert threshold used in the primary analysis seemed to be a better choice for clinician predictions.

DISCUSSION

The findings from this study showed that AI generally outperformed clinicians in retrospective patient outcome prediction, while the aggregation of predictions in the 7-clinician group enabled clinicians to outperform AI. In the prospective setting, subspecialized physicians outperformed AI except for delirium. Subspecialized physicians showed better prediction performance than physicians in training and nurses. In general, clinicians exhibited better performance in predicting mortality than AKI or delirium. Both clinicians and AI performed poorly in LOS prediction. The performance gap between clinicians and AI was the largest in retrospective delirium prediction. Limited inter-rater agreements within clinicians and between clinicians and AI showed large prediction variability and corroborated the performance gaps between clinicians and AI.

Clinicians performed better against AI in the prospective than the retrospective setting. This may have stemmed from that AI is better than clinicians in analyzing complex, multi-dimensional data, whereas clinicians have access to tacit patient information unavailable in, or unable to be

coded into, electronic medical records (EMR) which may contain important predictive information. Furthermore, the fact that clinicians knew the patients present in the ICU well in the prospective setting could have improved their predictions, in comparison with the retrospective setting where they knew the patients only through data. Therefore, the comparison between clinicians and AI in the prospective setting may not have been fair given the structural differences in available information. However, these differences inevitably exist in the real-world unless AI can observe patients in a more human-like manner.

In general, clinician-AI hybrid predictions outperformed clinicians or AI alone, in both the retrospective and prospective settings. In the retrospective classification tasks, the aggregation of seven clinician predictions still performed comparably or better than any hybrid approach. In both retrospective and prospective classification, the choice between the AND and OR operators should be driven by which performance metric is more important (sensitivity/NPV vs. specificity/PPV).

In recent years, there has been an increasing number of studies that compared clinicians and AI in various clinical tasks, particularly in diagnosis. The systematic reviews conducted by Takita et al.,¹¹ Salinas et al.,¹² and Shen et al.¹³ reviewed the studies that had compared clinicians and AI in medical diagnostics, skin cancer diagnosis, and disease diagnosis, respectively. In general, these reviews found that AI performed better or comparably to clinicians. They also noted that clinician performance increased with experience.

Notably, Goh et al.¹⁴ conducted a randomized clinical trial to investigate if the use of a large language model (LLM) improved diagnostic reasoning compared with conventional resources in a retrospective setting. They found that there was no difference between using the LLM and conventional resources. In fact, the best diagnostic performance resulted from the LLM alone.

In addition to diagnosis, AI has been shown to yield superior or comparable performance to clinicians in patient question answering, in terms of both response quality and empathy,¹⁵ symptom check,¹⁶ and triage.¹⁷ Furthermore, Huang et al.¹⁸ prospectively compared physician and nurse predictions of 28-day mortality in ICU patients. They found that physicians outperformed nurses by a substantial margin. Nurses with less than 10 years of ICU experience could not make accurate predictions.

Our findings are largely in line with the findings from these previous studies. Our novel contributions include: 1) comparison between retrospective and prospective settings; 2) focus on prognosis as opposed to diagnosis; and 3) investigation of a wide range of ICU patient outcomes.

Clinicians and AI demonstrated distinct but complementary strengths to patient outcome prediction in the ICU. Clinicians' better performance in the prospective setting speaks to the value of contextual bedside awareness. Experienced intensivists and aggregation of predictions from 7 clinicians showed how clinician prediction can be improved through clinical experience

and multiple clinicians complementing one another. In contrast, AI excelled when only complex, high-dimensional data were available in the retrospective setting.

While these speak to the prediction capabilities of clinicians and AI, several deployment issues need to be considered in practice. Based on our results, senior intensivists would be preferred to make predictions in real-world clinical settings. However, if only junior clinicians and nurses are present, aggregation of their predictions would also be effective. While AI may not be as good as senior intensivists, it can still provide more accurate predictions than junior clinicians and nurses in an automated manner. Given the complementary relationship between clinicians and AI's predictive abilities, combinations of clinician and AI predictions would be best if both are available, which is supported by the clinician-AI hybrid prediction results in this study.

Several limitations of this study should be considered. First, prospective data collection was hampered by the erroneous manual entry of patient identifiers resulting in approximately 35% of the clinician predictions failing data linkage. Second, prospective data collection occurred during the COVID pandemic, the unique situation of which may have affected clinical workload, decision-making, and patient characteristics. The AI models were trained on pre-COVID data and used to predict patients during the COVID pandemic. This may have led to decreased AI performance in the prospective setting. Ongoing data and performance drift monitoring and periodic model fine-tuning would be important since other seasonal (e.g., more trauma patients in summer as opposed to more viral respiratory tract infections in winter) and longer-term shifts (e.g., today's ICU patients have more co-morbidities compared to 25 years ago) in ICU patient characteristics also occur. Third, the retrospective cohort included only seven clinicians, which may limit generalizability. Fourth, in the prospective setting, the loosely defined clinician type grouping may not have fully captured the differences in clinical experience, training, and role. Fifth, prospective predictions were made within 16–24 hours of ICU admission, which may have disadvantaged clinicians compared with the AI model, which had access to all available data up to 24 hours. Sixth, the exclusion of the patients for AKI and delirium who did not have required measurements or already developed AKI or delirium during the first 24 hours in the ICU reduced the sample size by approximately 40% and introduced selection bias, as shown in Tables S2 and S3. Lastly, external validation is warranted since this study was conducted in a single Canadian province.

METHODS

Overall Study Design and Setting

Figure 5 shows the overall retrospective and prospective settings designed to compare the patient outcome prediction performances of ICU clinicians and AI. The retrospective and prospective comparisons involved 990 and 238 ICU admissions, respectively. AI models trained on 46,631 ICU admissions provided AI predictions in both settings.

This study was approved by the University of Calgary Conjoint Health Research Ethics Board (REB19-2107) and conducted in accordance with the Declaration of Helsinki. All clinician participants provided informed consent prior to participating in the study. Patient informed consent for the retrospective data pull was waived due to the large number of patients involved.

Predicted Patient Outcomes

In both retrospective and prospective settings, the primary patient outcomes were in-hospital mortality, 30-day post-discharge mortality, ICU LOS, and hospital LOS. Secondary outcomes were delirium and AKI during the ICU stay. To determine AKI outcomes, patients were required to have at least two serum creatinine measurements to allow assessment according to the Kidney Disease: Improving Global Outcomes (KDIGO) criteria,¹⁹ defined as an increase in serum creatinine of ≥ 0.3 mg/dL (≥ 26.5 μ mol/L) within 48 hours or an increase of ≥ 1.5 times the baseline value over 7 previous days. Delirium outcomes were determined using ICDSC, with scores ≥ 4 indicating delirium.

AI Model Development

The AI models used in both the retrospective and prospective settings were trained by applying machine learning to the ABeICU database.^{9,20} ABeICU includes 55,689 ICU admissions from 48,672 de-identified adult patients across 15 ICUs in Alberta, Canada, including tertiary-care and community ICUs caring for medical, surgical, trauma, burn, neuroscience, and vascular surgery patients, between February 2012 and December 2019. It integrates eCritical TRACER,²¹ an EMR-based ICU repository containing demographics, vital signs, laboratory results, acuity scores, clinical assessments, and intervention data, linked with the Vital Statistics database from Alberta Health for out-of-hospital mortality data.

The inclusion criteria applied to ABeICU were adult patients (18 years or older) with an ICU LOS longer than 24 hours. Also, 1,000 patients from December 2018 to June 2019 were reserved for retrospective comparisons, although data linkage issues led to 990 patients being used. For each outcome, patients with a missing target label were excluded. For AKI and delirium predictions, patients who had already developed AKI and delirium within the first 24 hours of ICU admission were excluded. The baseline characteristics of the included and excluded patients in AKI and delirium modeling were compared. The final model development cohort diagram for each predicted outcome is shown in Figure S10.

Each model was developed to use patient data from the first 24 hours in the ICU to predict a given outcome at 24 hours after ICU admission. Candidate features were selected based on the literature and data availability. All candidate features used for AI model development were also presented to clinicians in the retrospective setting to ensure both AI and clinicians used the same patient information to make predictions. Five models were trained for each classification outcome (in-hospital mortality, 30-day post-discharge mortality, AKI, and delirium): logistic regression, support vector machine, XGBoost, random forest, and neural network. In addition,

five regression models were trained to predict ICU LOS and hospital LOS: elastic net, support vector machine, XGBoost, random forest, and neural network.

Classification model performance was evaluated using sensitivity, specificity, accuracy, balanced accuracy, F1 score, the area under the precision-recall curve (AUPRC), and AUROC. Regression performance was assessed using MAE, RMSE, and the coefficient of determination (R^2).

All model training followed a standardized pipeline shown in Figure S11. Hyperparameter tuning, using the Bayesian optimization in Optuna, and best model selection optimized median AUROC and RMSE in 5-fold cross-validation for classification and regression, respectively. The hyperparameter space of each model is shown in Tables S25 and S26. Classification model output thresholds were optimized for F1 score in 5-fold cross-validation.

Furthermore, the calibration plot and Brier score of each classification model were generated, and the feature importance of all models was investigated using the SHAP analysis. Decision curve analysis was performed for each classification model as well in both the retrospective and prospective settings.

AI modeling and evaluation utilized scikit-learn, shap, dcurves, xgboost, and PyTorch in Python (version 3.9.19).

Retrospective Clinician Predictions

Three ICU physicians (two intensivists and one fellow) and four ICU nurses from the Foothills Medical Centre (FMC) in Calgary, Alberta, Canada were recruited to provide retrospective outcome predictions for the 990 ABeICU test set patients. Predictions were made based on de-identified patient data from the first 24 hours after ICU admission in secure REDCap web surveys. Patient data were presented in both tabular and visual formats, with selected variables (e.g., vital signs) displayed as time series plots to facilitate interpretation. An example REDCap survey is provided in the Supplementary Information.

Fifty from the 990 patient cases were randomly selected to be predicted by all seven clinicians. The remaining patients were evenly divided into five prediction sets, and each clinician was assigned to one set, containing approximately 190 patients. Each patient received predictions from one, two, or all seven clinicians. For analysis, the 990 patients were categorized into three groups based on the number of clinician predictions received (Table S27). Clinicians were given three weeks to predict their assigned cases. To minimize fatigue, clinicians were encouraged to distribute the workload across multiple sessions.

Prospective Clinician Predictions

In the prospective setting, we collected bedside predictions from clinicians at the FMC ICU, a 28-bed tertiary unit providing multidisciplinary care for medical, surgical, trauma, burn, vascular, and neuroscience patients. We included adult patients that had been in the ICU for 16-24 hours and asked clinicians to predict the primary and secondary outcomes for the patients they were

caring for. Clinicians were encouraged to use all information available in routine clinical practice. Clinician predictions were collected with a paper-based survey similar to the one used in the retrospective setting (see Supplementary Information), capturing clinician type, bed number, admission timestamps, and patient identifier to enable data linkage later for AI prediction and outcome determination.

In total, 465 predictions were collected for 287 patients admitted to the FMC ICU from September 1, 2020 to December 31, 2022, with each patient predicted by 1 to 4 clinicians. For analysis, clinicians were grouped based on their type: subspecialized physicians (attending physicians and fellows), physicians in training (residents and interns), and nurses (Table S28).

Retrospective and Prospective AI Predictions

For both the retrospective and prospective settings, the same best AI models predicted all primary and secondary outcomes for the patients that clinicians predicted, based on patient data from the first 24 hours in the ICU.

For the prospective setting, a data set comparable to ABeICU was extracted for the included patients to provide features (i.e., model input variables) for AI prediction and ground truth outcome labels for prediction performance evaluation. After data linkage between clinician prediction forms and the extracted data set, 238 ICU admissions from 215 patients remained. After further excluding outcomes that could not be determined or had already occurred during the first 24 hours in the ICU, the final numbers of predictions included in analysis ranged from 145 to 287 predictions for different outcomes (see Figure S12 for a cohort diagram).

Clinician-AI Hybrid Predictions

In both the retrospective and prospective settings, clinician and AI predictions were combined to study if such hybrid predictions outperformed clinician or AI predictions alone. Classification predictions were combined by applying logical AND and OR operators to binary clinician and AI predictions. Regression predictions were combined via averaging, minimum, and maximum. If there were multiple clinician predictions for the same patient, they were averaged first (and dichotomized for classification) before being combined with corresponding AI predictions.

Statistical Analyses

In both the retrospective and prospective settings, two-sample Wilcoxon rank-sum tests were used to compare the prediction performance metrics between the clinicians and AI based on 1,000 bootstrapped samples of the test data. These comparisons were stratified by the number of clinicians providing predictions and clinician type in the retrospective and prospective comparisons, respectively. The Wilcoxon rank-sum test assumes independence between the two groups, which should have been met since clinicians and AI made predictions independently. All predictions from the same clinician were treated as independent since they were for different ICU admissions.

Since the clinician predictions for the classification outcomes were recorded using a 5-point Likert scale, the top two options (i.e., “most likely” and “likely”) were combined to represent a positive prediction. When there were two or more clinicians predicting the same patient, their Likert-scale responses were averaged first and then dichotomized.

Furthermore, the prediction-level inter-rater agreement between clinicians and AI was assessed using Cohen’s κ and the intraclass correlation coefficient for the classification and regression predictions, respectively. The inter-rater agreement between clinicians when multiple clinicians predicted, in both the retrospective and prospective settings, was also assessed using the same methods.

All analyses were performed in Python (version 3.9.19).

Sensitivity Analysis

A sensitivity analysis on the Likert scale threshold was conducted by grouping “most likely”, “likely”, and “equally likely” to form a positive prediction. Prediction performance comparisons between clinicians and AI were repeated with this new threshold for the classification outcomes.

Data Availability

The ABeICU data cannot be shared without permission from the data custodians, Alberta Health Services and Alberta Health. The clinician predictions may be shared upon reasonable request without the corresponding patient data.

Code Availability

The source code of all data pre-processing, model training, and analyses is available on GitHub: https://github.com/data-intelligence-for-health-lab/Human_vs_AI_ICU_Patient_Outcome_Prediction.

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Author Contributions

CK performed all analyses, generated results, and wrote and revised the manuscript. CEV played a major role in the collection of both retrospective and prospective clinician predictions. HTS provided access to the ABeICU database, provided clinical oversight, critically analyzed the study design, and provided some clinician predictions. CBJ provided clinical and epidemiological/biostatistical oversight. JL conceived of and designed the study, provided resources and direct supervision to the research team, and wrote and revised the manuscript. All authors proofread and approved the final manuscript.

Competing Interests

JL is a co-founder and major shareholder of Symbiotic AI, Inc. The other authors do not have a competing interest.

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Figures

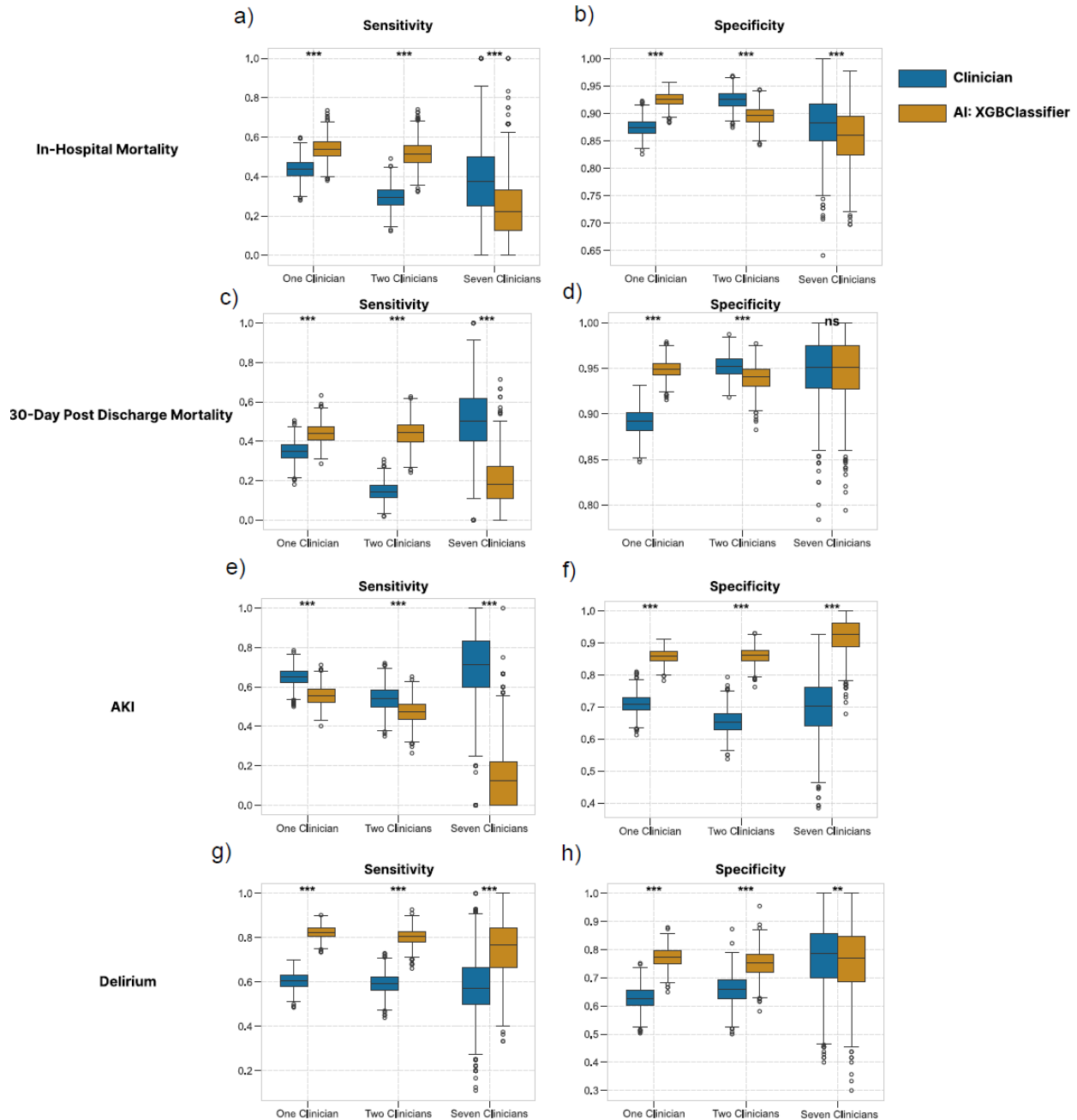


Figure 1. Retrospective comparisons between clinician and AI classification performances across different numbers of clinicians providing predictions. Boxplots show the 95% confidence intervals of clinician and AI prediction performances for: a) in-hospital mortality sensitivity, b) in-hospital mortality specificity, c) 30-day post-discharge mortality sensitivity, d) 30-day post-discharge mortality specificity, e) acute kidney injury (AKI) sensitivity, f) AKI specificity, g) delirium sensitivity, and h) delirium specificity. Comparisons are stratified by the number of clinicians contributing predictions: one, two, and seven clinicians. Statistically

significant differences between clinician and AI performances were evaluated using two-sample Wilcoxon rank-sum tests, with $P < .05$ (*), $P < .01$ (**), and $P < .001$ (***) indicating increasing levels of significance; “ns” indicates no statistically significant difference. Not all statistically significant differences would be clinically meaningful (e.g., delirium specificity in the seven-clinician group).

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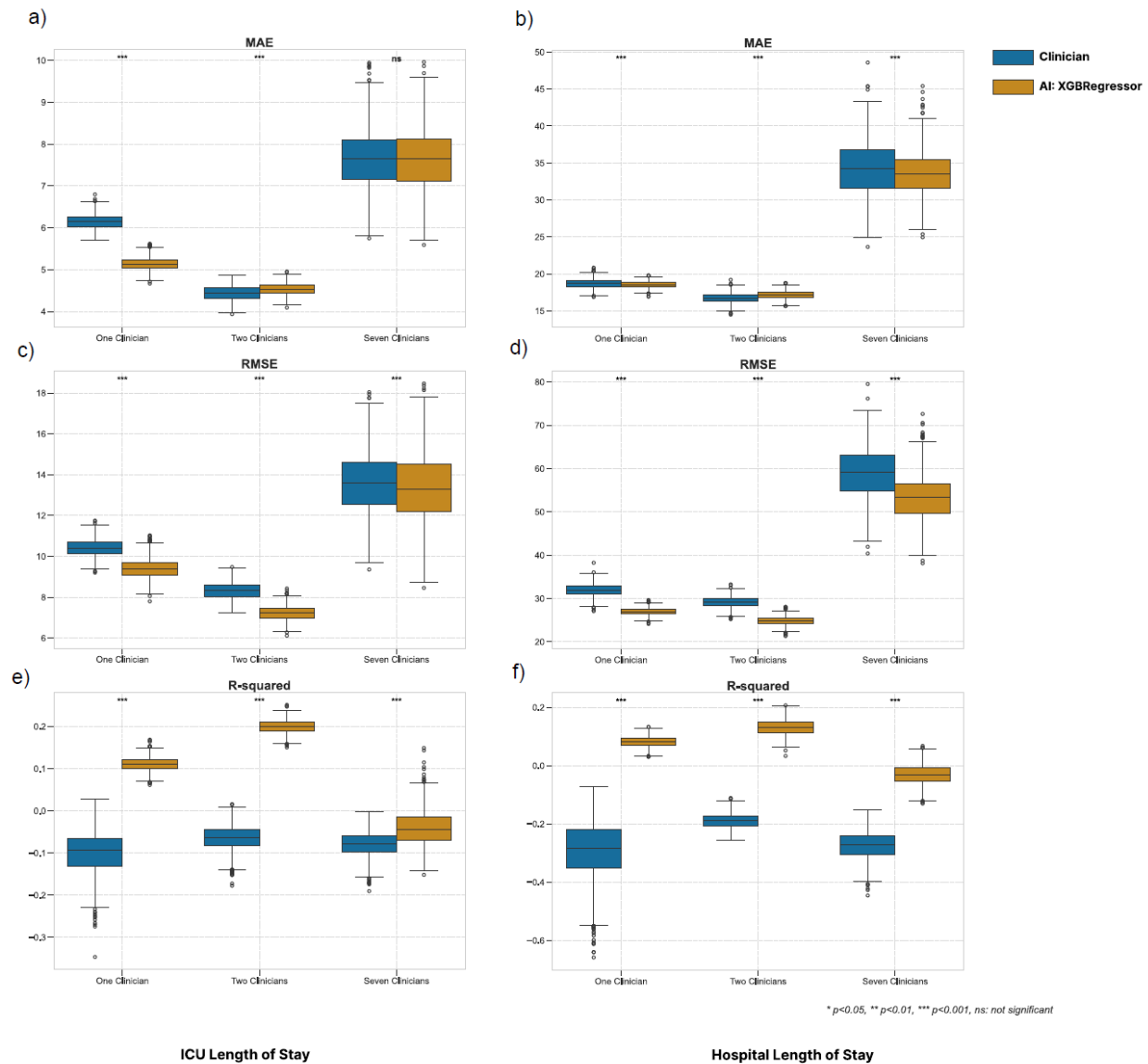


Figure 2. Retrospective comparisons between clinician and AI performances for length of stay predictions across different numbers of clinicians providing predictions. Boxplots show the 95% confidence intervals of clinician and AI prediction performances for: a) intensive care unit length of stay (ICU LOS) mean absolute error (MAE), b) hospital LOS MAE, c) ICU LOS root mean squared error (RMSE), d) hospital LOS RMSE, e) ICU LOS coefficient of determination (R^2), and f) hospital LOS R^2 . Comparisons are stratified by the number of clinicians contributing predictions: one, two, and seven clinicians. Statistically significant differences between clinician and AI performances were evaluated using two-sample Wilcoxon rank-sum tests, with $P < .05$ (*), $P < .01$ (**), and $P < .001$ (***) indicating increasing levels of significance; “ns” indicates no statistically significant difference. Not all statistically significant differences would be clinically meaningful (e.g., ICU length of stay RMSE in the seven-clinician group).

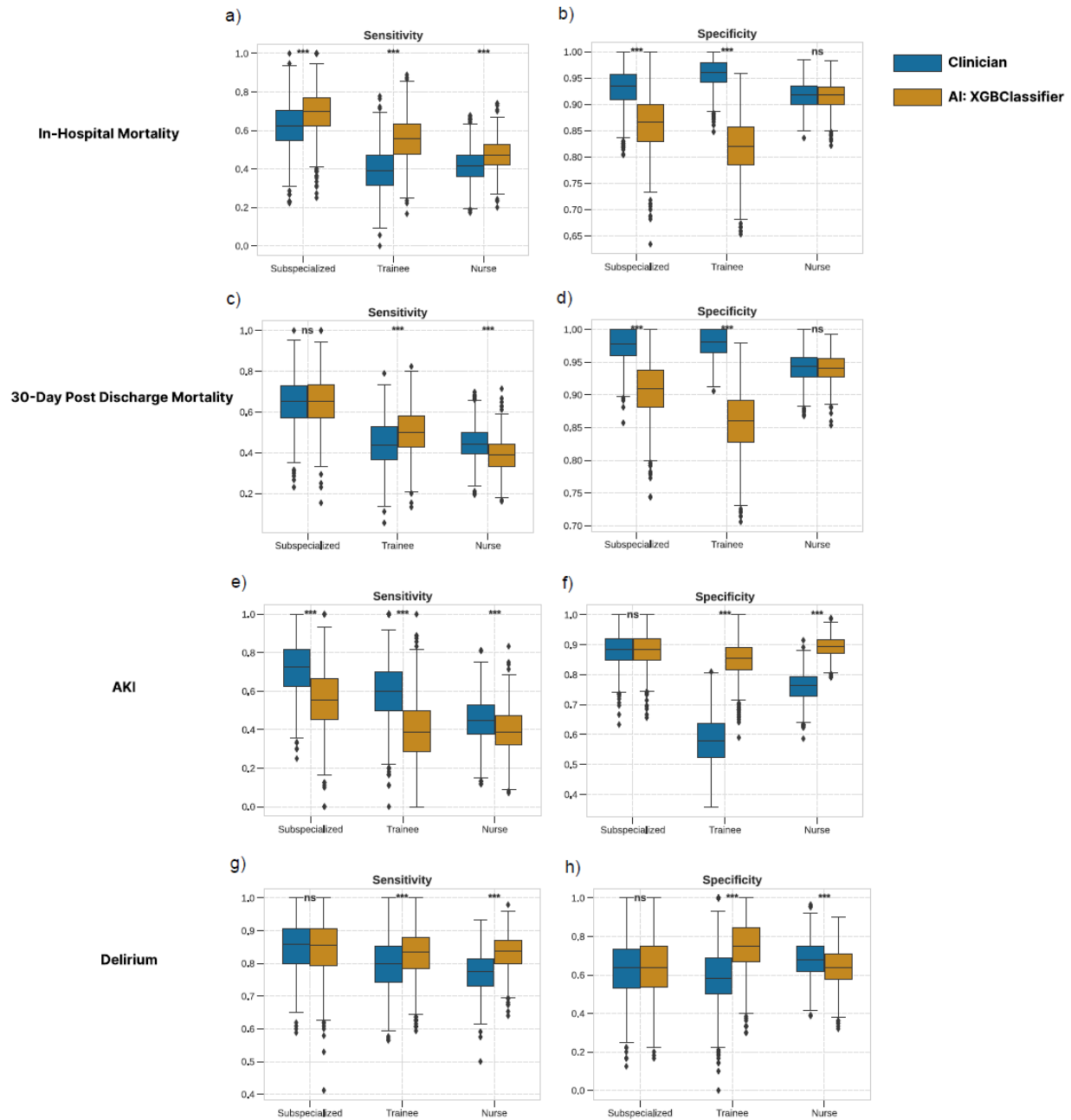


Figure 3. Prospective comparisons between clinician and AI classification performances across different clinician types. Boxplots show the 95% confidence intervals of clinician and AI prediction performances for: a) in-hospital mortality sensitivity, b) in-hospital mortality specificity, c) 30-day post-discharge mortality sensitivity, d) 30-day post-discharge mortality specificity, e) acute kidney injury (AKI) sensitivity, f) AKI specificity, g) delirium sensitivity, and h) delirium specificity. Comparisons are stratified by clinician type: subspecialized physicians (intensivists and fellows), physicians in training (residents and interns), and nurses. Statistically significant differences between clinician and AI performances were evaluated using two-sample Wilcoxon rank-sum tests, with $P < .05$ (*), $P < .01$ (**), and $P < .001$ (***) indicating

increasing levels of significance; “ns” indicates no statistically significant difference. Not all statistically significant differences would be clinically meaningful (e.g., delirium specificity in the nurse group).

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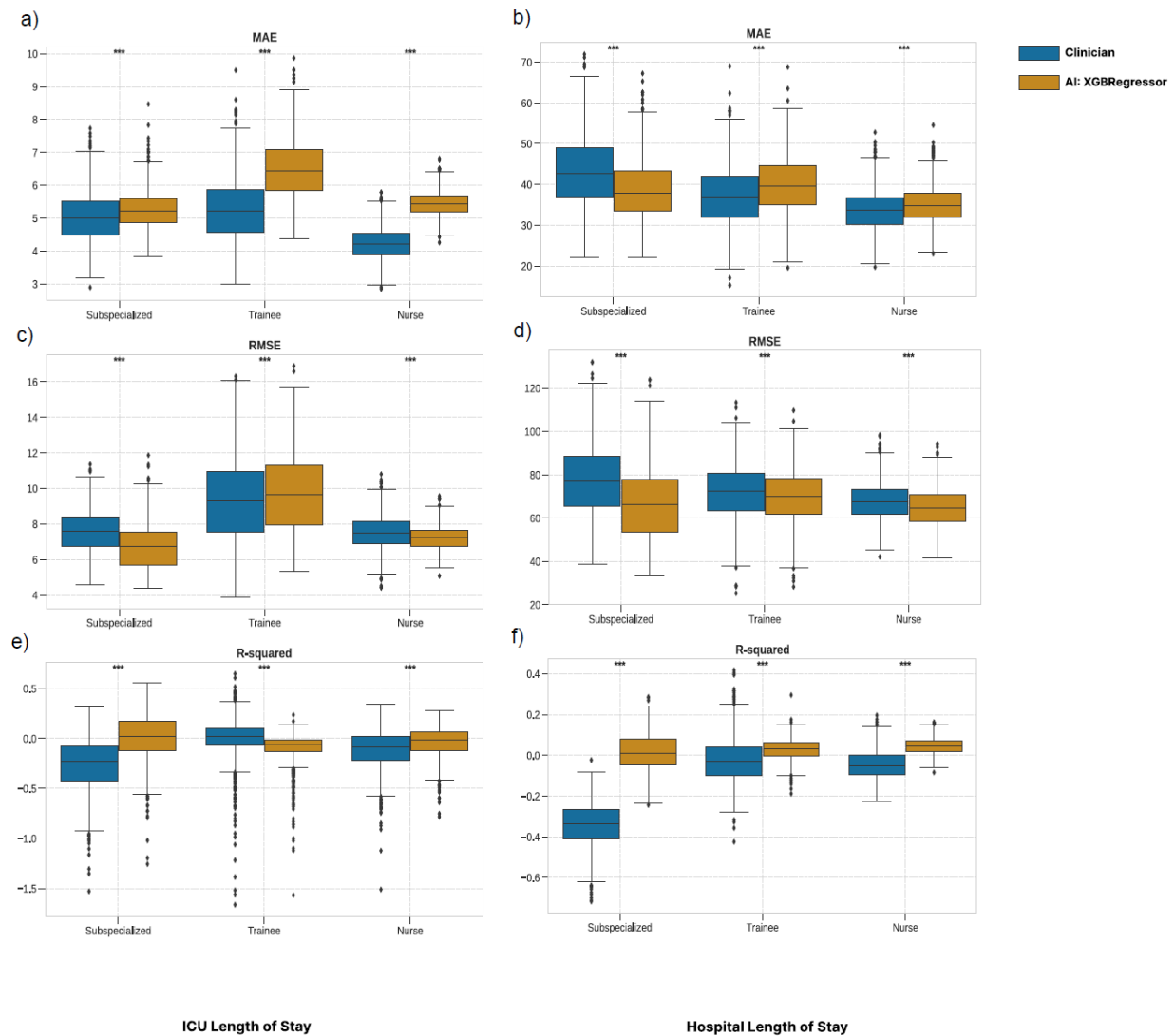


Figure 4. Prospective comparisons between clinician and AI performances for length of stay predictions across different clinician types. Boxplots show the 95% confidence intervals of clinician and AI prediction performances for: a) intensive care unit length of stay (ICU LOS) mean absolute error (MAE), b) hospital LOS MAE, c) ICU LOS root mean squared error (RMSE), d) hospital LOS RMSE, e) ICU LOS coefficient of determination (R^2), and f) hospital LOS R^2 . Comparisons are stratified by clinician type: subspecialized physicians (intensivists and fellows), physicians in training (residents and interns), and nurses. Statistically significant differences between clinician and AI performances were evaluated using two-sample Wilcoxon rank-sum tests, with $P < .05$ (*), $P < .01$ (**), and $P < .001$ (***) indicating increasing levels of significance. Not all statistically significant differences would be clinically meaningful (e.g., ICU length of stay RMSE in the nurse group).

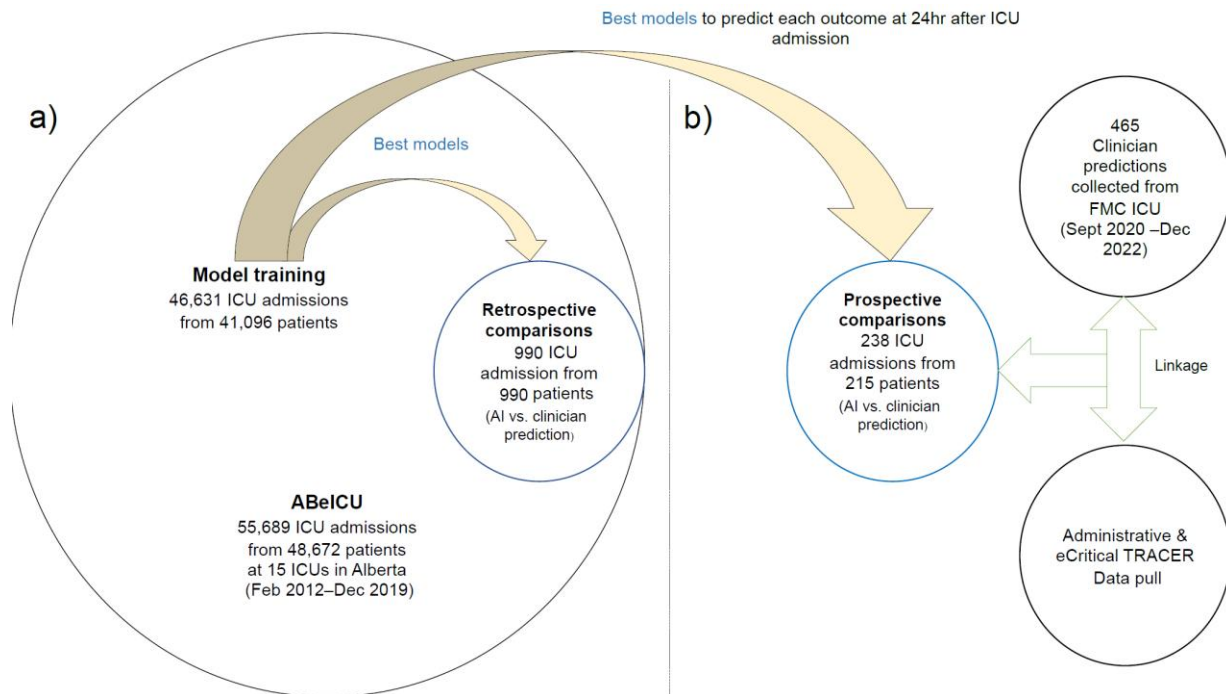


Figure 5. Study design and data flow for the retrospective and prospective comparisons of clinician and AI predictions. ABeICU was the ICU dataset used for AI training and retrospective comparisons. The best performing AI model for each outcome was selected and compared with clinician predictions in two settings: a) a retrospective setting where clinicians reviewed de-identified electronic medical records from the first 24 hours of 990 admissions from 990 unique patients in ABeICU to predict their outcomes, and b) a prospective setting where clinicians provided bedside predictions during actual patient care between 16 and 24 hours after ICU admission for 238 ICU admissions from 215 unique patients. The same AI models made predictions in both the retrospective and prospective settings based on patient data from the first 24 hours in the ICU.